

Parent Functions I: $\sqrt{x}$ and $|x|$ - Practice

Algebra II Honors | Unit 1 | Lesson 4 Practice | TEK A2.2.A

Objective: I can work with the piecewise definition of $|x|$, find domains of radical expressions, and reason about $\sqrt{x^2} = |x|$.

Section 1 — Recall

1. Evaluate $|x|$ at $x = -6, -2, 0, 5$.

2. Evaluate $\sqrt{x^2}$ and $|x|$ both at $x = -8$. What do you notice?

$\sqrt{64} =$ _____ $|-8| =$ _____

Section 2 — Apply

3. Give the domain, in interval notation, of $\sqrt{x-9}$, $\sqrt{5x+15}$, and $|x+9|$.

4. On $[4, 16]$, find the minimum and maximum of $\sqrt{x}$, and justify using the fact that $\sqrt{x}$ is always increasing.

min: _____ max: _____ Why: _____

Section 3 — Challenge

5. Solve $|2x - 6| = 10$, showing both cases.

$x =$ _____

6. Is $\sqrt{x^2} = |x|$ true for every real x , or only some? Prove it using both a positive and negative test value, and connect it to why $\sqrt{x}$ is called the "principal" root.

Section 4 — Explain It

7. A student claims that because $\sqrt{x^2} = |x|$ for all real x , it must also be true that $(\sqrt{x})^2 = x$ for all real x . Is the student correct? Explain why these two identities behave differently, addressing domain.
